EIDOS

ARC-AGI-3 grid world

abstract · turn-based

EXPANSE

NAO humanoid robot

physical · continuous

μ-GIL 2.0

A Spatial Abstraction Learner

Two worlds. One psyche. Nothing it did not earn.

A parsimonious instance of the Piagetian Modeler cognitive architecture, built to navigate the ARC-AGI-3 spatial reasoning corpus.

Michael S. P. Miller · SubThought Corporation · BICA-26, September 2026

The Piagetian Modeler

The architecture both systems implement. Constructivist neurosymbolic: mental models built solely through perception.

1

Perception is active and constructive

Sensory input passes through detectors into a variable-length TRAIT VECTOR, compared against stored cases. Miss → assimilate: mint a reifier, store a case, the ontology grows. Hit → accommodate: reinforce, spread activation. A categorical decision — no loss function, no gradient.

2

The world model is acquired, not given

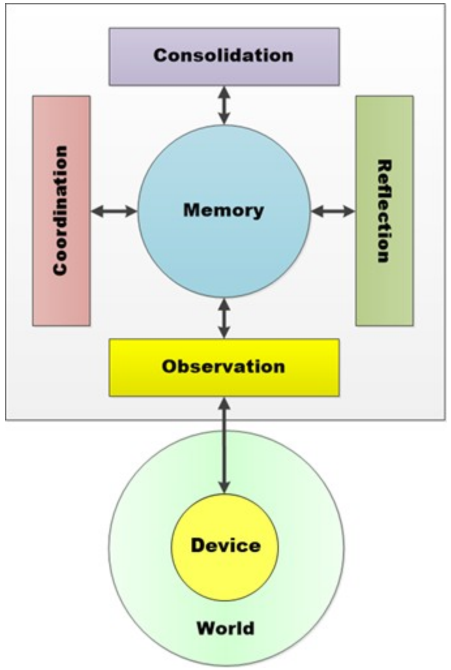
Every action produces proprioceptive feedback, compared against the prediction that preceded it. Confirmed reinforces; failed triggers compensation — the model is corrected at exactly the point of surprise.

3

One memory substrate, shared

Perception, coordination, reflection and consolidation read and write the same knowledge base. No separate training data, no separate inference model. Working memory IS long-term memory.

The Piagetian Modeler



Four processes, one Memory, one Device onto one World.

Miller, M. S. P. (2021). The Piagetian Modeler. AGI 2021, LNCS 13154, Springer. · GIL is the reference instantiation; μ-GIL is the second.

GIL — the reference implementation

~50 source files, over 20,000 lines of Premise. Four concurrent processes over one shared Totality.

Observation

`3 patterns`

Perceiver · Detectors · Matcher · Lexer · Storer · Activator · Imager (image schemas) · Locater / Mapper / Navigator

Coordination

`10 solvers`

Ameliorator (needs→objectives) · Deliberator · Reactor · Correlator/Regulator/Terminator · 136 reasoning operators, 7 strategies · Skemer · narrating

Reflection

`8 patterns`

motivation · simulation (daydreaming) · compensation · coping · discovery & experimentation · imitation · play — developmentally gated

Consolidation

`4 patterns`

compressing (chunking) · automating (skill formation) · forgetting (decay pruning) · recycling (resource reuse)

THE TOTALITY

six memory areas, one substrate

Control	Registry · Actor · Mind
Ontological	Schemes :M :R :A :E
Perceptual	icons · geons · lexemes
Associative	Cases · Trait index
Activative	8 realities · Glue
Imaginative	Canvas · Taxis · Route

μ-GIL — the parsimonious instance

~30 source files: 13 mechanism theories, 2 psyches, totality, imagination, portal. ~8,000 lines of Premise, 1,200 of Python.

SHARES

- the perception pipeline, faithfully
- assimilation / accommodation as the only two paths
- the seven-level spatial hierarchy, Scene through Map
- content-addressed reifiers — identity survives restarts
- the Correlator grading every consequence
- an audit trace at every pipeline stage

REPLACES

- 10 concurrent solvers → ~25 agents over one Agenda
- 136 reasoning operators → learned transitions only
- designed image schemas → discovered relational clusters
- 8 reflection patterns → simulation, exploration, discovery
- 4 consolidation patterns → Amneator and automaticity
- developmental gating, narration, theorizing → absent

INVENTS

- relational state layer — clusters, scale-normalised tokens
- anchor adoption — "this display can show what that plate shows"
- chain-walk — bounded BFS over the learned edge graph
- evidence confidence — majority-based edge recording
- the budget gate — learn the action budget from teleports
- question-the-map — accommodate the frontier ledger itself

The invented mechanisms are all responses to specific, diagnosed failures in live play — the kind a specification cannot contain.

Same architecture, two scales

No row is a different philosophy. Every µ-GIL entry is the GIL entry with what the task does not demand removed.

DIMENSION	GIL	µ-GIL
Source	~50 files, 20,000+ lines Premise	~30 files, ~8,000 Premise + 1,200 Python
Mechanisms	~60 agents, rules and services	~25 agents over one Agenda
Memory areas	6, incl. Glue spatial layout	6 — ~90% reused verbatim
Realities	8 parallel layers	4 — Observed, Desired, Expected, Imagined
Spatial ontology	7 levels, Scene through Map	the same 7 levels, contents learned
Planning	10 solvers, 136 reasoning ops	Navigator BFS over learned Action schemes
Forgetting	Amneator + 3 compressors	Amneator, decay-gated
Status	reference system, specified	runs live, wins level 1 reliably

Both answer the same question: how does a mind with no priors come to act competently in a world nobody described to it?

The Psyche — one contract, any world

A Psyche is the world interface — a sense organ. It mediates two flows and speaks exactly four tuples.

The four tuples

PERCEPT *inward*

`:Modality :Channel :Address :Data :Content :Moment`

ATTEMPT *outward*

`:Action :Parameters :Token :By`

RESULT *inward*

`:Action :Status :Reason :Moment :Token`

URGE *inward*

`:Need :Source :Delta :Moment :Token`

Interoception, proprioception, perception and recollection are four sequences over the same pipeline.

What varies, what does not

INVARIANT — the mind

- the tuple grammar and slot names
- Perceiver · Matcher · Lexer · Storer · Activator
- Scener · Simulator · Coordinator · Correlator
- every downstream stage works on the trait set alone

PER-DEVICE — the world

- :Channel — which sensory domain exists at all
- :Data — which parsable structures arrive
- :Content — the device's own slot vocabulary
- :Action — the device's actuation repertoire

World one — the Eidos

εἶδος, "to see." The device is the ARC-AGI-3 interactive reasoning benchmark. LS20, "Locksmith."

THE DEVICE

A 64 × 64 grid, 16 colours, 7 standardised actions, turn-based — the world changes only when the agent acts. No instructions are given: mechanics, goal and win condition must all be discovered. Scored by RHAE = (human actions ÷ agent actions)^2.

PERCEPTS · one channel: Grid

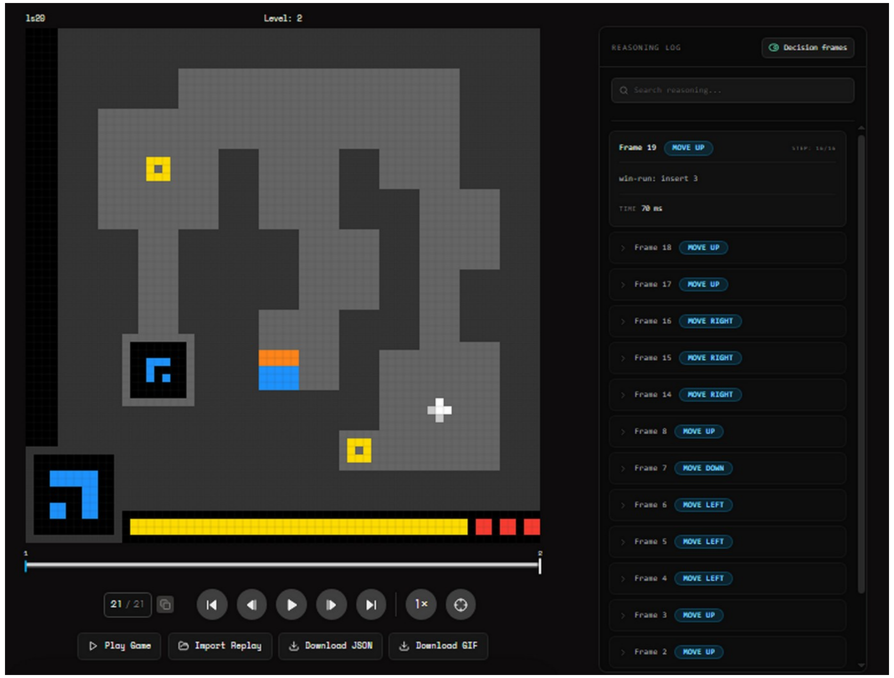
`grid-state`	full board, level, state, available actions
`grid-delta`	only what changed, keyed by causing action
`game-event`	win or game-over transition

ACTUATIONS · seven

`RESET · ACTION1–4 (up/down/left/right) · ACTION5 (interact, select, rotate) · ACTION6 (click at x,y) · ACTION7 (undo)`

Connected components become traits — region 9:3x3:5@35,11.
Nothing in the mind knows what a lock, a key or a door is.

LS20 "LOCKSMITH" · the live board and its reasoning log



Every action's basis rides to the API's reasoning field — the competition's own log records WHY.

World two — the Expanse

Latin expansum, "spread out." The device is a NAO humanoid: 25 degrees of freedom, in a room.

THE DEVICE

SoftBank NAO over NAOqi or ROS 2. Two head cameras, a four-microphone array, torso IMU, sonar, bumpers, capacitive head sensors, 25 joints. Continuous, noisy — and it does not wait for the mind to finish thinking.

PERCEPTS · eight channels

`Visual`	camera-frame · depth-map · object-detected
`Auditory`	audio-sample · transcription · sound-event
`Haptic`	contact · bumper · collision
`Proprioceptive`	joint-state · imu-reading · battery-state
`Spatial`	odometry · pose · occupancy-grid
`Communication`	speech-heard · gesture-seen · touch-signal

ACTUATIONS · 35 in the spec

Locomotion	`walk · turn · stop · stand · sit · navigate`
Gaze	`look · track`
Manipulation	`reach · grasp · release · lift · place`
Gesture	`point · wave · nod · shake-head`
Communication	`speak · emote · display · set-led`
Motor	`joint-move · pose-move · trajectory`
System	`e-stop · set-mode · capture-photo`

HOMEOSTATIC URGES

power (battery < 15%) · thermal · balance (IMU tilt > 0.3 rad) · safety (collision) · connectivity

A body has needs a grid does not. Urges are how they reach the mind.

One interface, two devices

The tuple is identical. Every slot inside it is the device speaking — and detection is the only world-specific stage.

EIDOS

```
[PERCEPT :Modality Eidos    :Channel Grid            :Data grid-state
          :Content {:Grid {…4096 ints…} :State playing :LevelsCompleted 0
                    :ActionsUsed 12 :HumanActions 8 :AvailableActions {…}}]
```

EXPANSE

```
[PERCEPT :Modality Expanse  :Channel Proprioceptive  :Data joint-state
          :Content {:Joints {{:Name "LKneePitch" :Position 0.8 :Effort 3.1} …}
                    :Unit "rad"}]
```

Matcher · Lexer · Storer · Scener · Activator · Simulator · Coordinator · Correlator operate on the trait set and nothing else. They cannot tell a grid from a knee. Add a world by adding a detector clause.

One spatial hierarchy, two worlds

Scene through Map — the same seven levels, whether the body is a grid piece or a humanoid in a room.

LEVEL	EIDOS	EXPANSE
Map	the whole game — every level	the site
Region	a quadrant of a level	a building
Area	a corridor system	a floor
Place	an arrangement — the position fingerprint	a spot: by the door
Locale	a chamber of the maze	the kitchen
Venue	the board	the room
Scene	one frame's regions	one camera field

The grammar is shared. The contents are learned.

- **Locater**
  builds bottom-up from perceived containment and adjacency
- **Mapper**
  accumulates top-down as new groupings are met
- **Navigator**
  BFS over the graph; mints a Route of waypoints

Why this closes the horizon

A forty-step excursion at PLACE granularity is beyond any bounded walk. The same journey is three hops at LOCALE granularity. The level-2 wall was not a missing route — it was a missing level of abstraction.

An independent parallel in vision theory

Blumberg's Neural Rendering describes the same epistemic structure μ-GIL's core loop implements.

THE FORMALISM · Blumberg (2026)

$$\hat{y}_{world}, \hat{y}_{body} = F(s_{world}, s_{body}, a) \quad \text{(i)}$$

a forward model predicts world and body consequences

$$\delta = \text{encode}(y) - \hat{y} \quad \text{(ii)}$$

the structured residual: predicted versus observed

$$s_{t+1} = G(s_t, \delta, a) \quad \text{(iii)}$$

an empirically specified gate routes the residual

THE CORRESPONDENCE · in μ-GIL

F — the TransitionScheme row — action + context → predicted moved / vanished / appeared

δ — the Correlator's symmetric difference over predicted and observed arrangement keys

G — the Correlator / Regulator / Terminator chain — reinforce, or overwrite at the point of surprise

The whitelist that separates the controllable piece from HUD churn is μ-GIL's separation of body state from world state.

NAPOT · receive → transform → re-express

Multiple partial projections converge at a receiver, the receiver integrates, and the result is re-expressed downstream. In μ-GIL the Matcher's Jaccard scan is this topology — traits converge at each Case, overlap is the integration, the 85% threshold is the quorum gate, the winner is re-expressed. So is the band auction (winner-take-all) and the chain-walk (value propagating back through a learned graph).

The WHAT and WHERE are present.
The WHEN is missing.

Blumberg, M. (2026). Embodied Neural Rendering in Self-Aware Networks. Self Aware Networks Institute. DOI: 10.5281/zenodo.21331180

Results — 45 live runs against the competition server

Up to 16 chained 8-turn sessions per run, the Totality saved and restored across every session boundary.

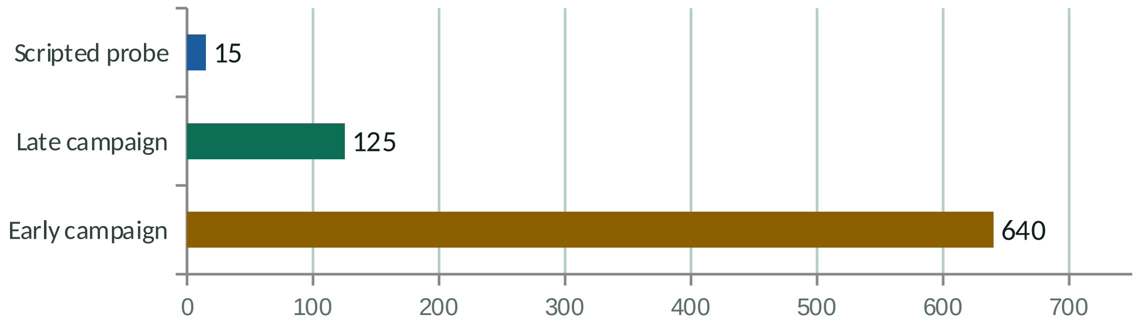
11
first live runs completed zero levels

12
the run that first completed level 1

80%+
level-1 win rate from run 14 onward

0
level-2 completions, in any run

Cost per level-1 win



The gap between 125 and 15 is the price of building the abstractions rather than being given them.

The visible signature of learning

- **Frontier burn-down**
  plateaus at 30 in run 1; burns to 22 by run 7
- **Assimilation ratio**
  share of arrivals at arrangements seen before — climbs faster
- **Exploration rate**
  drops to zero: the mind has learned enough structure to exploit it
- **Model confirmation**
  surprises front-loaded, decaying as the world model converges

The boundary it drew

Level 2 was never completed. This is the most useful result in the campaign.

THE ENERGY ECONOMY

~21 actions per life, then the piece teleports home and the display reverts

~19 actions for the winning route — spawn, to the cross, to the portal

5 muse probes tested it; the furthest reached the cross, none entered the portal

THE SHAPE OF THE WALL

The winning mechanism sat 59 walk-steps away, inside the budget, and the mind could not see it. One-step dreams all read known-and-burned; the chain-walk found no frontier to route to; question-the-map won by default and took two-thirds of the level-2 action budget re-verifying a map that was already fine.

The diagnosis

What is missing is not compute or budget. The mind was planning at PLACE granularity — forty single steps, whose value no one-step dream and no bounded walk can see. It had no LOCALE above them, so it could not ask the short question. Evidence-driven exploration has a horizon, and forty-five runs found exactly where a flat map ends.

Plateaus break on representation, never on iteration.

µ-GIL 2.0 — what changes

Tabula rasa, no imported direction, and decisioning carried by concurrent agents over the shared Totality.

Out — **Imported waypoint routes**

The one lever that moved level 2 — and the one thing the mind could not have earned.

Out — **Single-function, ranked selection**

Memoryless by construction: every tick re-evaluated from scratch, so no commitment survived the tick that produced it.

In — **Agents over a shared Agenda**

Proposers enable their act in Desired; gates impede; convergent proposals accumulate. The Deliberator selects the most-activated act, and Cronos decays what nobody re-endorses.

In — **The Action scheme as the unit**

Context + Goals → Result, wired so activation flows through the slots. An empty :Result IS the frontier; a bred composite IS a multi-step plan.

In — **A Critic, and failure as a monad**

Rehearse in Imagined and promote only what satisfies the telon; a failure mints its dyad, an impediment, and an agenda item to find the cause.

None of these is a new idea. Roughly 90% of GIL's Totality — Agenda, Attempt, Activation, the Scheme machinery — is reusable verbatim.

The long-horizon problem

In 2022 the longest task a frontier model could finish alone was seconds of human-equivalent work. Today's agents sustain hours — and the doubling interval is itself shrinking.

H1

intra-context

Many coupled steps in one window. Reasoning that verifies and recovers before the chain derails.

H2

cross-context

The task outgrows the window. State compressed, externalised, checkpointed, faithfully resumed.

H3

cross-task

An open-ended stream with shifting goals. Reusable skills that accumulate over time.

Mid-run verification, natively

Inline judging is where research is ahead of product. µ-GIL cannot act without first dreaming the outcome and settling it confirmed or surprised.

Durable state, tested to destruction

Sessions end every eight turns by design. Save, wipe, reload, resume — the H2 problem faced 45 times on the public record.

A real reward signal

A live competition API grades every action, and every decision's basis rides to its reasoning field. No simulated environment, no verifier to reward-hack.

METR, Time Horizon 1.1 · Sequoia, "2026: This is AGI" · Dong et al., Towards Long-Horizon Agents: A Survey (July 2026)

The honest frontier

A mind that earns its knowledge is slow to start — and impossible to fool about what it does not know.

Proved

A minimal Piagetian mind wins a real ARC-AGI-3 level reliably from cold, learning its spatial ontology from scratch, with every decision auditable.

Bounded

Evidence-driven exploration has a horizon. Forty-five live runs found it precisely, and never crossed it without imported direction.

Next

Close the horizon from inside — then make the same core carry a NAO body.

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YouTube — youtube.com/@CognitiveArchitectures

Building Sentient Beings — Miller & Blumberg (2025) · zenodo.org/records/15522356

Building Minds with Patterns — Miller (2018) · available on Amazon

Coding Artificial Minds — Miller (2026) · forthcoming

µ-GIL: A Spatial Abstraction Learner — BICA-26 proceedings, September 2026